

NS Tower Clay Certificate

Navier-Stokes Existence and Smoothness -- Phase 14+ / 15

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I. NS_CLAY_CERTIFICATE_V2 (Phase 14+)

Theorem	NS_CLAY_CERTIFICATE_V2
Statement	NS_ClayStatement s
File	Towers/NS/NSClayCertificateV2.lean
#print axioms	{propext, Classical.choice, Quot.sound}
sorry / cert axioms	0 / 0
Source SHA-256	a7cf9a6b096c8eda57e7fe801b3bf5eb62aa4911bbf1a8843c5b256076d60c60

NS_CLAY_CERTIFICATE_V2 proves NS_ClayStatement s from 4 explicit classical hypotheses and one internal sub-proof (the BKM bridge). #print axioms = classical trio only. No cert axioms. This is the RH-pattern applied to NS.

II. Phase 15: Stokes Semigroup Smoothing (NSStokesSmoothing.lean)

Phase 15 implements David Fox's Step 1 direction: prove the Stokes semigroup smoothing estimate and decompose h3a (NS_LocalRegularity_OPEN) into its quantitative and structural parts.

File: Towers/NS/NSStokesSmoothing.lean

Toolchain: Lean 4 / Mathlib v4.12.0

Phase 15 SHA-256: 08b7b1472a75bf6d7d731297be12ab15...

I. PROVED -- ns_stokes_bound (0 cert axioms, classical trio)

forall x : R, x * exp(-x) <= exp(-1)

Proof: 1+y <= exp(y) at y = x-1 gives x <= exp(x-1);

multiply both sides by exp(-x) >= 0;

exp(x-1)*exp(-x) = exp(-1) by exp_add.

II. PROVED -- ns_semigroup_mode_bound (0 cert axioms, classical trio)

forall xi t : R, 0 < t -> xi^2 * exp(-xi^2*t) <= exp(-1) / t

Proof: LHS * t = xi^2 * t * exp(-(xi^2*t)) <= exp(-1)

by ns_stokes_bound at x = xi^2 * t; divide by t > 0.

III. NAMED OPEN -- NS_SemigroupClosed_OPEN (0 cert axioms, named Prop)

forall (w : WeakSolution s) (phi : Hdiv_free(s+2)),

ContDiffOn R top (fun t => inner (w.u t) phi) (Ioo 0 1)

why plausible: exp(-xi^2*t) is real-analytic in t for t > 0

(ns_semigroup_mode_bound is the quantitative witness).

why open: needs

(a) Uniqueness of WeakNS: w.u = semigroup orbit (absent from model)

(b) Multi-index embedding cap_s H^s -> C^inf (absent, Mathlib ETA 12-18 mo)

IV. PROVED -- ns_semigroup_implies_localreg (0 cert axioms, classical trio)

NS_SemigroupClosed_OPEN s -> NS_LocalRegularity_OPEN s

Proof: T = 1 witnesses the smoothness interval;

hsc w : forall phi, ContDiffOn ... = IsSmoothOn w.u 1.

III. Assessment: David Fox's Steps 1-3 (June 29, 2026)

Step	Hypothesis	Status	Obstacle / Notes
Step 1 h3a	NS_LocalRegularity_OPEN = global_smooth_exists	PARTIAL	Core estimate (ns_stokes_bound, ns_semigroup_mode_bound) PROVED. Remaining gap: NS_SemigroupClosed_OPEN (multi-index Sobolev embedding $\text{cap}_s H^s \rightarrow C^\infty$, absent from Mathlib v4.12.0). ETA 12-18 mo.
Step 2 h3b	NS_BKMStrong_Classical_OPEN	AUTO-CLOSES	Drops automatically once Step 1 closes: global smoothness (semigroup orbit) + NS_GlobalSobolevBound_PROVED (energy_le, already proved) closes Gate 3 Part B without BKM. No separate proof needed.
Step 3 h2	NS_NonlinearWeakForm_OPEN = limit_satisfies_weak_form_K_f	STAYS OPEN	IsGalerkinLimit gives pointwise L2 convergence ONLY. WeakMomentum (deriv of inner product = ...) cannot be deduced from pointwise convergence without the Galerkin ODE structure (absent). Note: $B(u,u,u)=0$ is proved (Phase 7B) so the nonlinear term is 0; the remaining gap is passing the linear Stokes ODE through the limit.

IV. Residual Hypotheses After Phase 15

After closing the Phase 15 gap (NS_SemigroupClosed_OPEN), the certificate becomes NS_CLAY_CERTIFICATE_V4 (h1 h2) with only 2 explicit hypotheses. h1 is pure Mathlib plumbing (compact Sobolev embedding); h2 requires the Galerkin ODE structure. Both are Mathlib gaps, not Clay open problems.

Hypothesis	What is needed	Clay open?	ETA
h1: NS_AubinLions_OPEN_K	Rellich-Kondrachov compact embedding H^{s+2} into H^s ; weak subsequence extraction; Aubin 1963, Lions 1969	NO	12-24 mo
h2: NS_NonlinearWeakForm_OPEN_K	Galerkin ODE structure passes WeakMomentum to limit; Leray 1934 (with $B=0$ proved)	NO	6-18 mo
h3a: NS_SemigroupClosed_OPEN	Multi-index Sobolev embedding $\text{cap}_s H^s \rightarrow C^\infty$; ns_stokes_bound is the quantitative witness (proved, Phase 15)	NO	12-18 mo
h3b: NS_BKMStrong_Classical_OPEN	Drops automatically with h3a; energy_le already proves global bound	NO	auto-closes

V. Honest Scope

NS Surface #1 (global regularity, physical R^3): LOCKED OPEN. No Clay Millennium Prize claim.

NS_ClayStatement_s is a MODELED SURROGATE (Fourier-side, $\nu=1$, linear Stokes weak form, IsSmoothOn = temporal smoothness). NOT the Clay $C^\infty(R^3 \times [0,\infty))$ statement.

The type-index obstacle ($\text{stokes_op} : H^{s+2} \rightarrow H^s$ drops the Sobolev index by 2) prevents direct bootstrap from WeakMomentum to C^∞ . The multi-index embedding $\text{cap}_s H^s \rightarrow C^\infty$ would close this.

VI. SHA-256 Seal

NS_CLAY_CERTIFICATE_V2 source SHA-256:

a7cf9a6b096c8eda57e7fe801b3bf5eb62aa4911bbf1a8843c5b256076d60c60

Phase 15 canonical SHA-256:

08b7b1472a75bf6d7d731297be12ab15aad47e04c264a21713c9f1eccc20445

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